

ASTROPATH: Sky-View Autonomous Drone Navigation and Road Damage Detection System

Author & Owner: Mallu Diswardhan Reddy | **Date:** June 2026 | **Designation:** ECE Systems Engineering Portfolio

Email: 438malludiswardhanreddy@gmail.com | **GitHub:** @438malludiswardhanreddy-sketch

ABSTRACT

The name **ASTROPATH** is derived from **ASTRO** (meaning Sky) and **PATH** (meaning View or Way), representing a sky-view system for autonomous aerial infrastructure monitoring and road damage diagnostics. Urban road infrastructure maintenance remains a significant challenge for municipality corporations in developing economies. Conventional manual inspection techniques are slow, error-prone, and hazardous. This research presents ASTROPATH, an end-to-end, edge-assisted, autonomous aerial system designed to automate road damage diagnostics. Incorporating a custom 3D Kalman Filter for sensor fusion, a dual-stage neural network framework (YOLOv4-tiny and MobileNetV2) for edge-based classification, and a dynamic safety-priority route optimisation engine, ASTROPATH provides continuous pothole mapping and rapid response coordination. The system runs real-time perception pipelines on Raspberry Pi and ESP32-CAM hardware interfaces, syncing telemetry data with a centralised municipal dashboard to streamline public work repairs.

1. Objective

The primary objectives of the ASTROPATH system are:

- **Autonomous Damage Detection:** Automating the identification and severity analysis of potholes and road cracks using high-speed edge devices.
- **Reliable Geolocation Mapping:** Projecting pixel-level computer vision detections into precise real-world GPS coordinates using drone altitude, attitude (gimbal pitch/roll), and telemetry.
- **Multi-Sensor State Estimation:** Filtering noisy GPS, IMU, and barometric altimeter data through a custom 3D Kalman Filter to ensure stable flight path tracing.
- **Dynamic Path Planning and Safety Overrides:** Implementing collision avoidance algorithms (for birds, trees, and low-hanging wires) and autonomous safety triggers (Return-to-Home and Emergency Landing) during adverse wind, rain, or low battery conditions.
- **Decentralised Citizen Reporting:** Integrating public citizen upload pipelines to verify and complement aerial survey maps.

2. Methodology

2.1 System Architecture

ASTROPATH is divided into three key architectural layers:

1. **Perception Layer (Edge Device):** Executes object detection and classification. It processes video frames from onboard camera streams (via RTSP or local WiFi interfaces like ESP32-CAM) and estimates the severity of detected hazards.
2. **Navigation and Fusion Layer (Flight Controller):** Acquires spatial coordinates from the GPS module and IMU. A 3D Kalman Filter fuses these streams to produce high-accuracy state estimations.
3. **Orchestration Layer (Cloud/Municipal Server):** Exposes RESTful API endpoints for telemetry logs and citizen uploads, maintaining a centralised SQLite database and dynamic Flask dashboard for visualising pothole maps.

2.2 Sensor Fusion (3D Kalman Filter)

Drone navigation modules rely on sensor fusion to counteract high Gaussian noise in raw GPS signals (~5m variance) and altimeter sensors (~30cm variance). The state vector is defined as:

State Vector: $x = [\text{lat}, \text{lon}, \text{alt}]^T$

The prediction and correction loops update the covariance matrix P against process noise Q and measurement noise covariance R :

1. Predict State:

$$P_{k|k-1} = P_{k-1|k-1} + Q$$

2. Correct State (GPS & Altimeter Ingestion):

$$\begin{aligned} K &= P_{k|k-1} (P_{k|k-1} + R)^{-1} \\ x_k &= x_{k|k-1} + K (z_k - x_{k|k-1}) \\ P_{k|k} &= (I - K) P_{k|k-1} \end{aligned}$$

2.3 Computer Vision Perception Layer

- **YOLOv4-tiny Model:** Used to scan video frames at 30 FPS. It localises candidates for road anomalies (specifically bounding box coordinates).
- **MobileNetV2 Binary Classifier:** Acts as a secondary verification stage. It runs deep feature extraction on cropped bounding box images to confirm if they represent potholes or false positives (like shadows or dry leaves).
- **Severity Estimation:** Calculated based on the ratio of bounding box area to overall frame size. Low, Medium, and High classifications dictate the urgency level logged in the database.

3. Run-Time Results (Simulation Verification)

The system was validated using a high-fidelity end-to-end mission simulator, tracing a path across Solapur, Maharashtra, India (coordinates starting at 17.6599, 75.9064). Below is the structured log representing the state overrides and actions:

Step	Mode	Trigger / Hazard	Threat Score	System Action / Override
1	TAKEOFF	Initial GPS Acquisition	0.00	Hold altitude at 15.0m. Fuse signals.
3	CRUISE	Tree branch blocking path	0.70	Lateral bypass detour (+4m East).
5	CRUISE	Low-hanging power line	0.80	Vertical bypass detour (+6m climb).
6	CRUISE	Wind gust @ 28.5 kts	Critical	Throttled speed to 1.5 m/s (Stabilisation).
7	CRUISE	Bird flock in flight vector	0.55	Hover-in-place state activated.
8	CRUISE	Heavy rain @ 18.0 mm/h	Warning	Flagged precipitation safety warning.
9	CRUISE	Onboard battery @ 12.0%	Critical	Trigger RTH. Land at safe LZ: (17.6639, 75.9093).

4. Future Scope

- **Real-time Edge TPU Integration:** Deploying compiled Google Coral Edge TPU models to accelerate MobileNetV2 inference latency to under 5 ms.
- **Autonomous Swarm Mapping:** Coordinating multiple edge-enabled micro-UAVs to divide large municipal sectors for rapid grid mapping.
- **Advanced Depth/Thermal Vision:** Integrating stereo cameras to estimate pothole depth and volume directly from flight, allowing accurate material estimation for public repair teams.
- **5G Telemetry Streams:** Incorporating high-bandwidth 5G transceivers to stream raw visual feeds back to control centres when GPU offloading is necessary.

Mallu Diswardhan Reddy

Project Inventor & Systems Architect

ASTROPATH Aerial Autonomous Systems Initiative